

EDUCATION

Princeton Quantum Initiative, Princeton University

Princeton, USA

Ph.D. in Quantum Science

2024-2029 (*expected*)

- Advisors: Prof. Dmitry Abanin, Prof. Sarang Gopalakrishnan
- GPA: 4.0/4.0
- Princeton First Year Fellowship in Natural Science & Engineering

2024-25

IIT Bombay

Mumbai, India

Bachelors in Electrical Engineering

2020 - 2024

- Minor in Physics
- Advisors: Prof. Sai V, Prof. Bhaskaran M, Prof. Kasturi S
- GPA: 9.77/10
- *Awards*

- Graduated with Honors, top 3% of class 2024
- Undergraduate Research Distinction 2024
- Dhananjay Fellowship 2023
- Chanakya Quantum Fellowship 2023
- Institute Academic Prize 2023
- Undergraduate Research Award 2023

RESEARCH INTERESTS

quantum error correction, quantum simulation, quantum metrology, tensor networks

PUBLICATIONS

1. SM, D. Abanin “Local autonomous inference machines for quantum LDPC codes”
[arXiv:2609.38139]
2. A. Basu, P. Kothari, SM “Sharp bounds on the ground state energy of the SYK model”
[arXiv:2607.27185]
3. SM, Y. Zhang, D. Malz, D. Abanin, S. Gopalakrishnan “Algorithmic locality via provable convergence in quantum tensor networks ” [arXiv:2604.21919]
4. SM*, G. Sommers*, J. Tindall, D. Abanin “Belief Propagation & Tensor Network Expansions for Many-Body Quantum Systems: Rigorous Results and Fundamental Limits”
[arXiv:2604.03228]
5. SM*, Y. Zhang* “Beyond Belief Propagation: Cluster-Corrected Tensor Network Contraction with Exponential Convergence”
[arXiv:2510.02290, PRX Quantum 7, 033010, TQC 2026]
6. SM, S. Gopalakrishnan “Metrology of open quantum systems from emitted radiation”
[arXiv:2504.13815, PRA Letter 112, L060601]
7. K. Agaram, SM, V. Garg “Preparing stabilizer states via path-aware reinforcement learning” [Neurips Machine Learning for Physical Sciences, APS Global Summit 2025]
8. A. Arora*, SM*, A. Zyuzin, P. Hakonen, B. Muralidharan,
“Steady-state dynamics and correlations in quantum-dot Cooper pair splitters”
[arXiv:2406.06053, npj Quantum Information 11, 40 (2025), APS March Meeting 2024]
9. SM, K. Jana, B. Muralidharan,
“Are Symmetry Protected Topological Phases Immune to Dephasing? — A topological electronics perspective” [arXiv:2305.11149, J. Phys. D: Appl. Phys. 57 135308]
10. SM, M. Parashar, A. Bhatla, D. Broadway, J.P. Tetienne, K. Saha,
“Optimized Current Density Reconstruction from Widefield Quantum Diamond Magnetic Field Maps” [arXiv:2402.17781, Phys. Rev. Applied 22, 014015]

11. SM, R. Singh, K. Gharavi, J. Baugh, B. Muralidharan,
 “On the microscopics of proximity effects in one-dimensional superconducting hybrid
 systems” [[arXiv:2411.12733](#)] *denotes equal contribution

AWARDS	• DAAD-WISE Fellowship, German Academic Exchange Service	2023
	• MITACS Globalink Fellowship, Canada (unavailed)	2023
	• Aalto Science Fellowship, Aalto University	2022
	• All India Rank 150, the JEE Advance exam (top 0.09%)	2020
	• KVPY Award, Govt. of India (unavailed)	2020

TALKS	1. Beyond Belief Propagation: Cluster-Corrected Tensor Network Contraction	
	(a) Google Quantum AI Tensor Network Meeting (online) ♦	2025
	(b) Thompson group meeting, Princeton, USA ♦	2025
	(c) International Conference on Theory of Quantum Computation, Communication and Cryptography (TQC)	2026
	2. Taming the many-body Hilbert Space: Tensor Networks and Belief Propagation	
	(a) Physics Department Seminar, IIT Bombay, India ♦	2026
	(b) Algorithms and Complexity Seminar, Princeton, USA ♦	2026
	(c) ITensor Seminar, Flatiron Institute, NYC, USA ♦	2026
	(d) Max-Planck Quantum Theory Group Seminar ♦(online)	2026
	3. Algorithmic locality in quantum tensor networks	
	(a) Interplay between neural and tensor networks, Grenoble, France	2026
	4. Local autonomous inference machines for quantum LDPC codes	
	(a) Princeton Quantum Initiative Group Meeting, Princeton, NJ	2026
	5. Quantum metrology of continuously monitored open systems	
	(a) Blackboard seminar, UC San Diego, USA	2025
	(b) Quantum Engineering Seminar, IIT Bombay, India ♦	2026
	(c) Relativistic Quantum Information Online Circuit	2026
	6. Solid-state entanglement generation with Cooper-pair splitters	
	(a) APS march meeting, Minneapolis, MN, USA	2024
♦ denotes invited talk		

EXPERIENCE	Undergraduate researcher	
	<i>TU Munich & MPQ, Germany.</i>	2023.05 - 2023.07
	Undergraduate researcher	
	<i>Aalto University, Finland.</i>	2022.05 - 2022.07
	Research internship (virtual)	
	<i>QIntern.</i>	2022.05 - 2022.07

MEDIA	Quantum Zeitgeist “Cluster-corrected Tensor Network Contraction Achieves Exponential Convergence with Rigorous Error Bounds”
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TEACHING	1. Advanced Quantum Mechanics, Princeton	2026
	2. Information theory and applications, Princeton	2025
	3. Quantum information and computing, IIT Bombay	2024
	4. Calculus I, IIT Bombay	2022

5. Complex Analysis, IIT Bombay 2022
6. Differential equations, IIT Bombay 2021
7. Linear algebra, IIT Bombay 2021
8. Calculus II, IIT Bombay 2021, 2022

SKILLS

Programming: Python, Julia

Libraries: QuTiP, Qiskit, ITensor, Stim, PyTorch

Analysis: Mathematical proofs, optimization, information theory, statistical & data analysis

MENTORSHIP

Mentored Krishna Agaram, junior undergraduate at IIT Bombay

Institute Student Mentor: to twelve freshmen at IIT Bombay

Department Academic Mentor: to eight sophomores at IIT Bombay

SERVICE

Journal referee Physical Review Letters, Physical Review A

Conference referee for NeurIPS machine learning for physical sciences 2024, 2025

Quantum outreach

1. Mentored 12+ junior undergraduates in quantum computing research at IIT Bombay
[[Learning \(with\) quantum computers](#), [Quantum machine learning](#)]

2. Princeton Undergrads in Quantum Talk [A conversation on quantum information](#)

Co-organizer of the quantum information and computing group at IIT Bombay 2023-24

Co-organizer of the [quantum error correction journal club at Princeton](#) 2025-

Co-organizer of the [Princeton many-body reading group](#) 2026-